

# The KCP Limited

Industrial Noise

# Key Topics

- Compensation Aspects
- Properties of Sound
- Risk Criteria
- Sound Measuring Instruments
- Sound Surveys
- Noise Control Programs
- Industrial Audiometry
- Exposure Regulations

# Definitions

- *Occupational Hearing Loss*
  - Hearing impairment in one or both ears that results from one's employment
- *Acoustic Trauma*
  - Injury to the sensorineural elements of the inner ear

# Compensation Aspects

- Approximately 16.9% of workers exposed to environments of 85 dBA and above
- Some state worker compensation laws cover hearing loss
- 1.7 million workers aged 50–59 years have compensable hearing loss
- Employers are compelled to address noise in the workplace

# Compensation Aspects (cont.)

- Evaluating Noise-Induced Hearing Loss (NIHL)
  - employee history of hearing loss
  - occupational history
  - results of audiometry exam
  - results of otological exam
  - ruling out nonindustrial causes

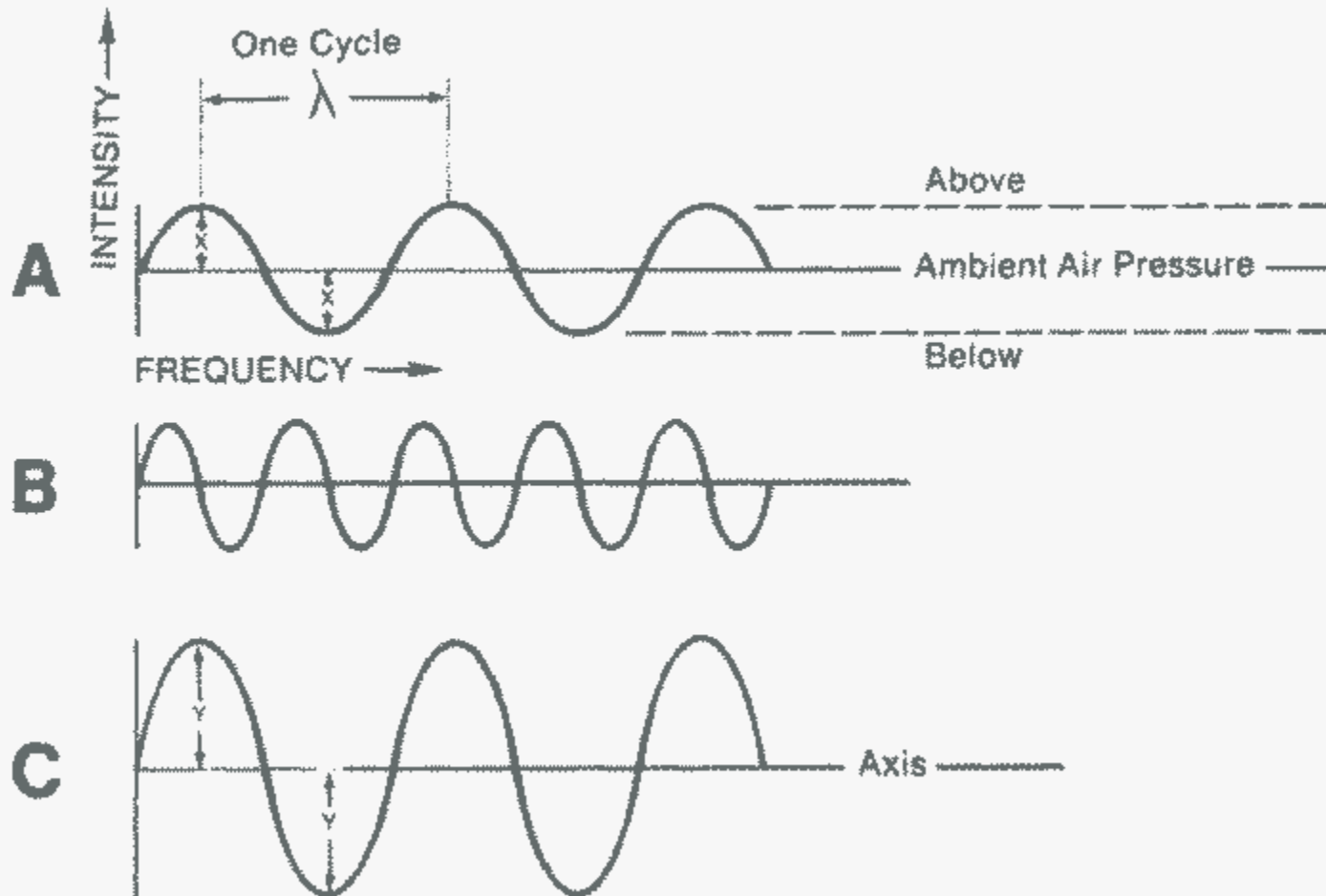
# Properties of Sound

- Noise = unwanted sound
  - interferes with speech and warning signals
  - auditory fatigue
  - hearing damage
  - annoyance

# Properties of Sound (cont.)

- Sound—propagation of compressions and rarefactions in air
  - Frequency
    - Number of times a vibrating body completes one cycle of motion
    - Interpreted as pitch
    - Human hearing range 20–20,000 Hz
  - Wavelength
    - Distance between two successive points on a wave
    - Important property in designing sound control measures

# Properties of Sound (cont.)





# Properties of Sound (cont.)

- Sound Power
  - Amount of energy per unit time radiating from a sound source
  - Measured in watts
  - $L_w = 10 \log \frac{W}{W_0}$ 
    - $W$  = sound power in watts
    - $W_0$  = reference sound power ( $10^{-12}$  watts)

# Properties of Sound (cont.)

- Sound Pressure Level
  - Measured as the root mean square
  - Preferred unit is the decibel
  - Referenced to a sound pressure of 20  $\mu\text{Pa}$
  - $L_p = 20 \log \frac{p}{p_o}$ 
    - $p$  = measured root mean square sound pressure
    - $p_o$  = reference sound pressure (20  $\mu\text{Pa}$ )

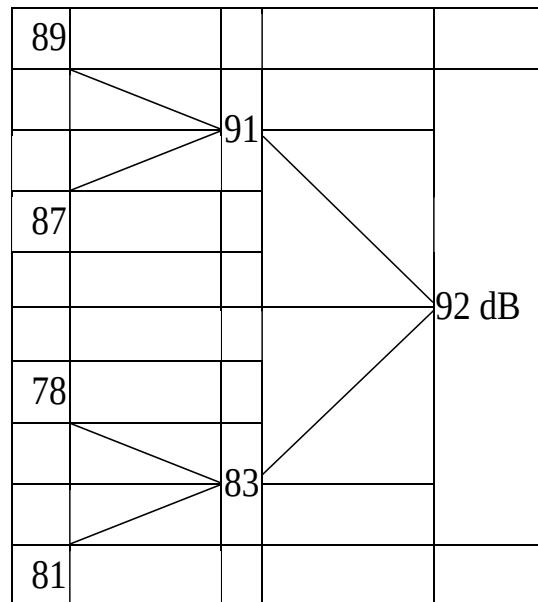
# Properties of Sound (cont.)

- Adding SPLs of Different Noise Sources

| Difference in<br>Decibel Values | Add to<br>Higher Value |
|---------------------------------|------------------------|
| 0 or 1 dB                       | 3 dB                   |
| 2 or 3 dB                       | 2 dB                   |
| 4 to 9 dB                       | 1 dB                   |
| 10 dB or more                   | 0 dB                   |

# Properties of Sound (cont.)

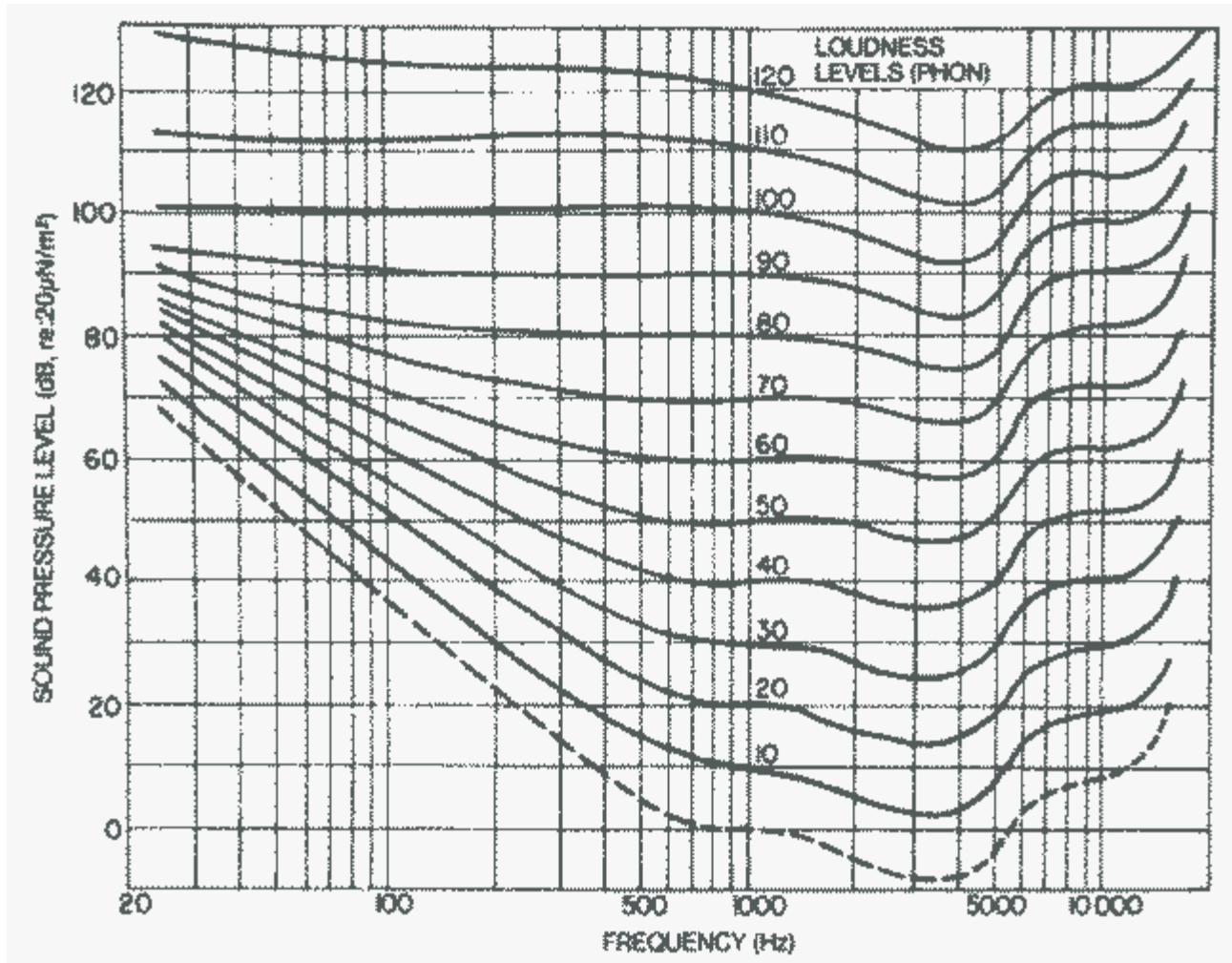
- When measured at the same location, four noise sources have sound pressure levels of 89, 87, 78, and 81 dB, respectively. What would the sound pressure level at this location be if all four sources were running concurrently?



# Properties of Sound (cont.)

- Loudness
  - Represents the subjective response to sound pressure and frequency
  - Lower frequencies are attenuated, higher frequencies accentuated
  - The nonlinear response of the human ear is represented by equal loudness contours

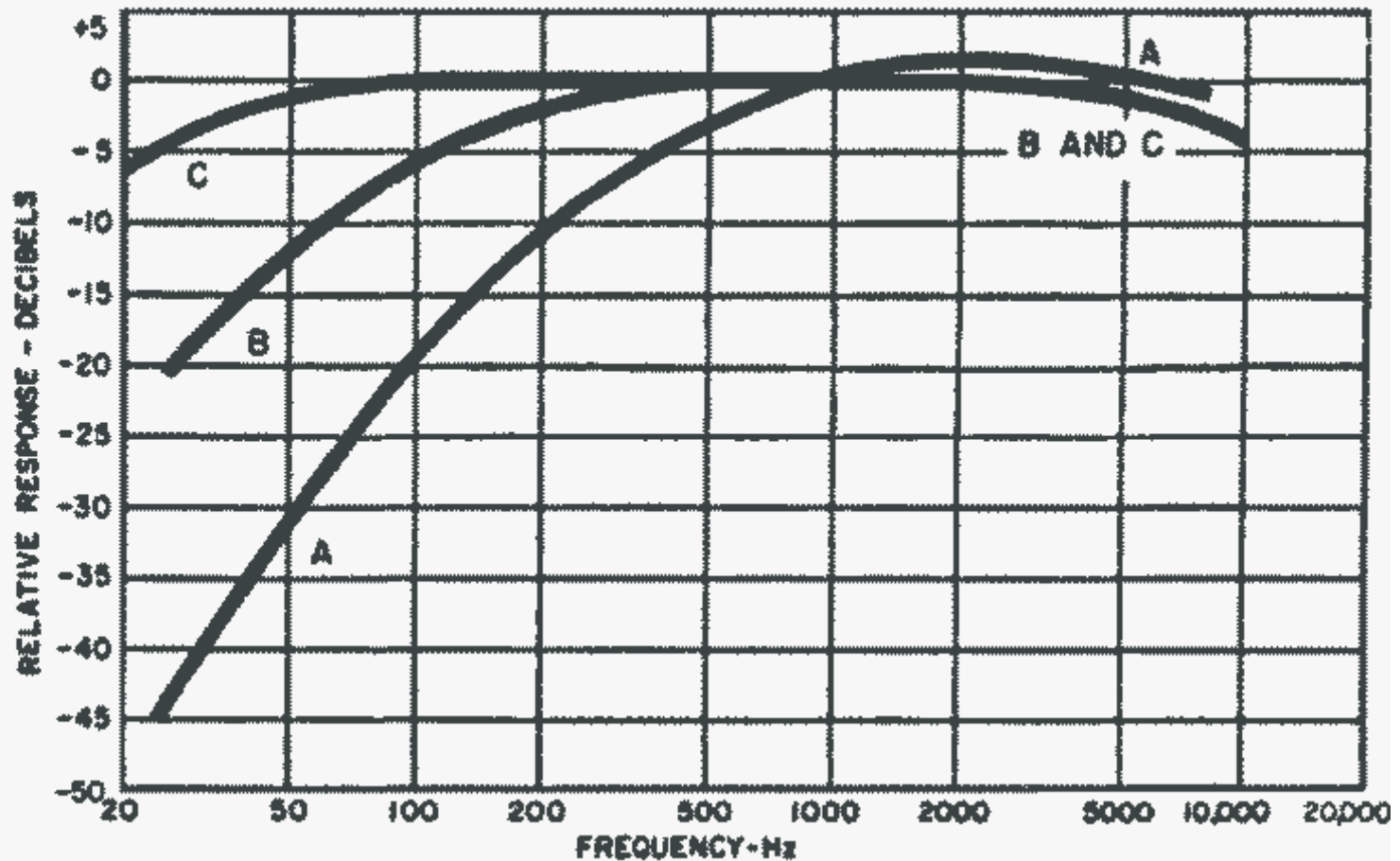
# Properties of Sound (cont.)



# Properties of Sound (cont.)

- Sound Pressure Weighting
  - Three (3) weighting networks: A, B, and C
  - A approximates the response of the human ear at low sound pressure levels
    - Low frequencies are attenuated
  - C was designed for high sound pressure levels
  - Sound level meters generally measure A- and C-weighted levels

# Properties of Sound (cont.)





# Occupational Damage Risk Criteria

- Evaluating Hearing Ability
  - Measure the ability to hear speech or to understand speech
- AMA Guide
  - Hearing thresholds measured at 500, 1000, 2000, and 3000 Hz
- Risk Factors for NIHL
  - Noise intensity
  - Frequency spectrum
  - Total work duration
  - Age of worker
  - Additional factors

# Sound Measuring Instruments

- Sound Level Meter
  - general purpose device for measuring sound pressure levels at fixed locations
  - microphone, amp, weighted network response, and indicating meter
  - A and C weighted networks
  - slow and fast response
  - generally measure between 40–140 dB

# Sound Measuring Instruments (cont.)

- Octave Band Analyzer
  - for evaluating the frequency spectrum of sound
  - generally cover nine octave bands ranging from 22–11,300 Hz
  - useful for determining appropriate engineering noise controls
    - low- and high-frequency noise require different control methods

# Sound Measuring Instruments (cont.)

- Noise Dosimeter
  - worn by worker across the workday
  - calculates a time-weighted average noise exposure
  - used to measure compliance with the OSHA Noise PEL

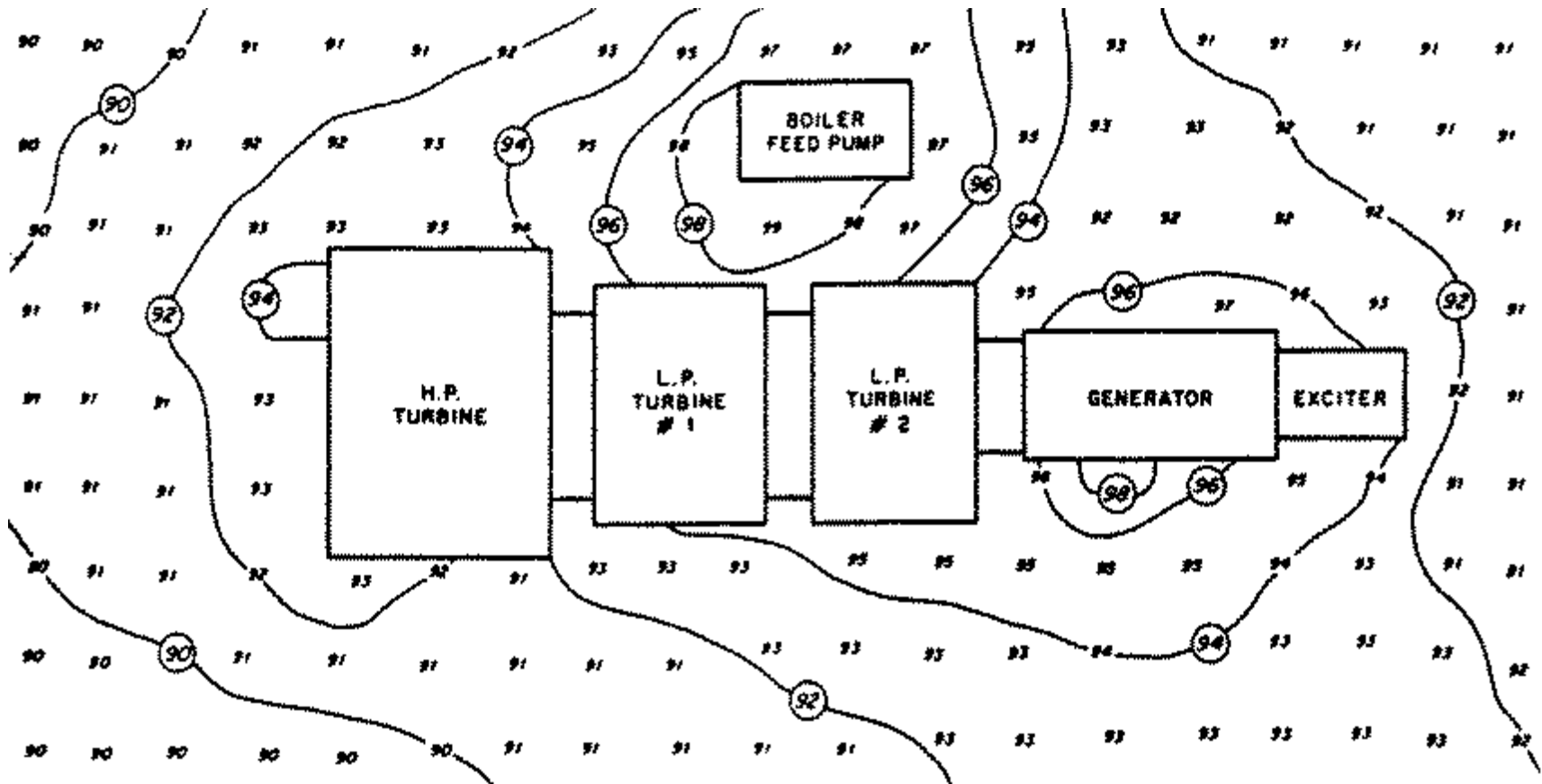
# Sound Surveys

- Purpose
  - to assess noise levels at work stations
  - to develop appropriate engineering or administrative controls
  - to define areas where hearing protection is needed
  - to identify work areas where workers may require audiometric testing

# Sound Surveys (cont.)

- Detailed Noise Survey Elements
  - area measurements
    - center of work area
    - sound level contours
  - workstation measurements
    - to evaluate worker exposure
  - exposure duration
    - for workers exposed to noise levels of 85 dBA and above

# Sound Surveys (cont.)



# Sound Surveys

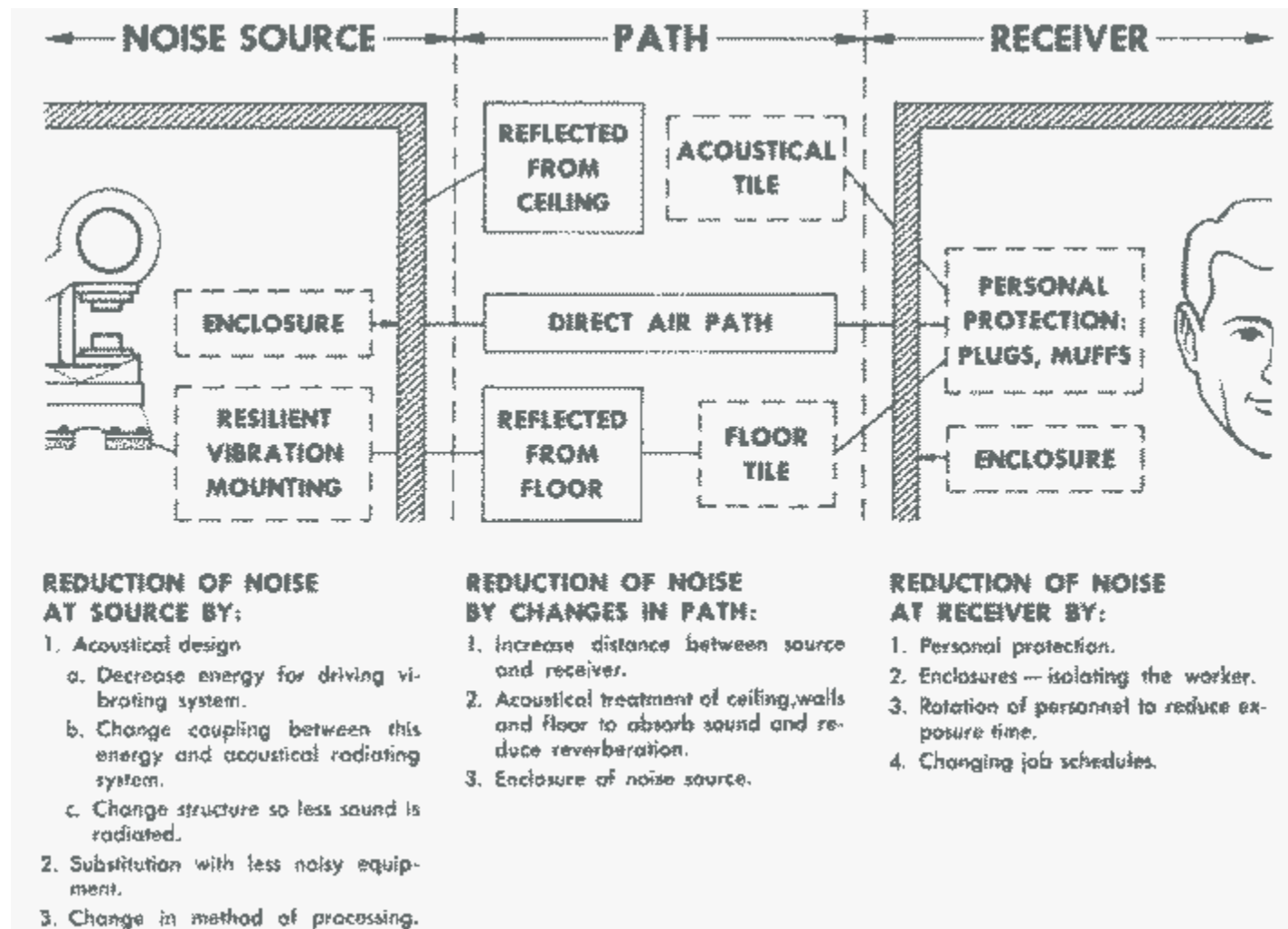
- Classes of Noise Exposure
  - *continuous noise*
    - broadband noise of constant pressure and frequency level
  - *intermittent noise*
    - exposure to broadband sound pressure level several times during a normal workday
  - *impact noise*
    - noise that is less than one-half second in duration



# Noise Control Programs

- Types of Controls
  - engineering
  - administrative
  - personal hearing protection

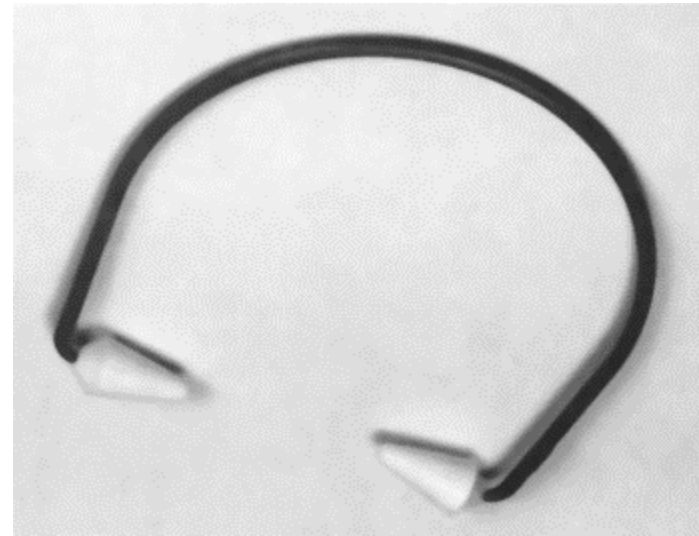
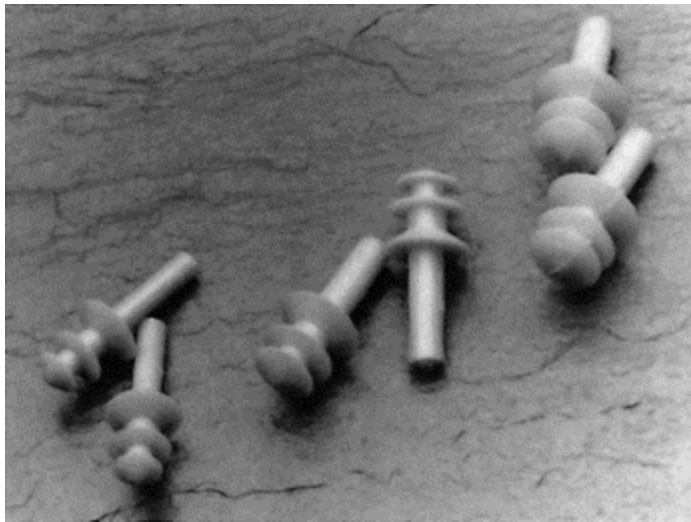
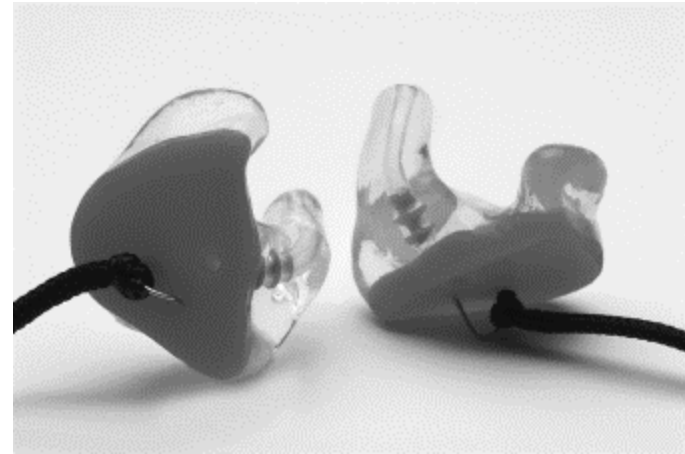
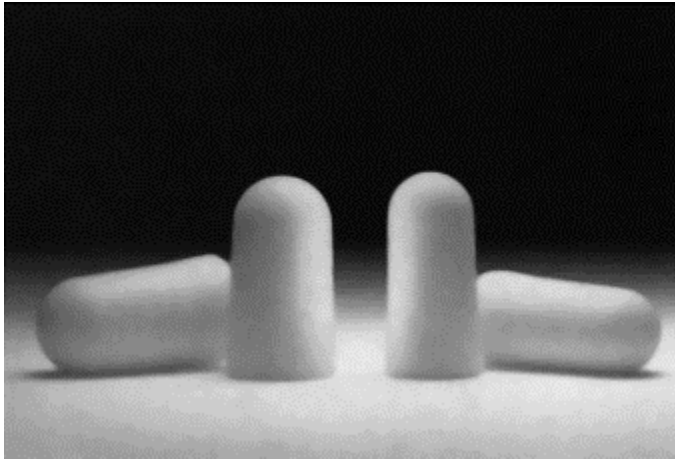
# Noise Control Programs (cont.)



# Noise Control

- Classes of Hearing Protection
  - enclosures
  - aural inserts or earplugs
  - supraaural protectors or canal caps
  - circumaural protectors or earmuffs

# Noise Control (cont.)



# Noise Control (cont.)

- Factors in Selecting Hearing Protection
  - attenuation factors
    - Noise Reduction Rating (NRR)
  - work environment and tasks
  - frequency and duration of noise exposure
  - whether communication with other workers is required

# Industrial Audiometry

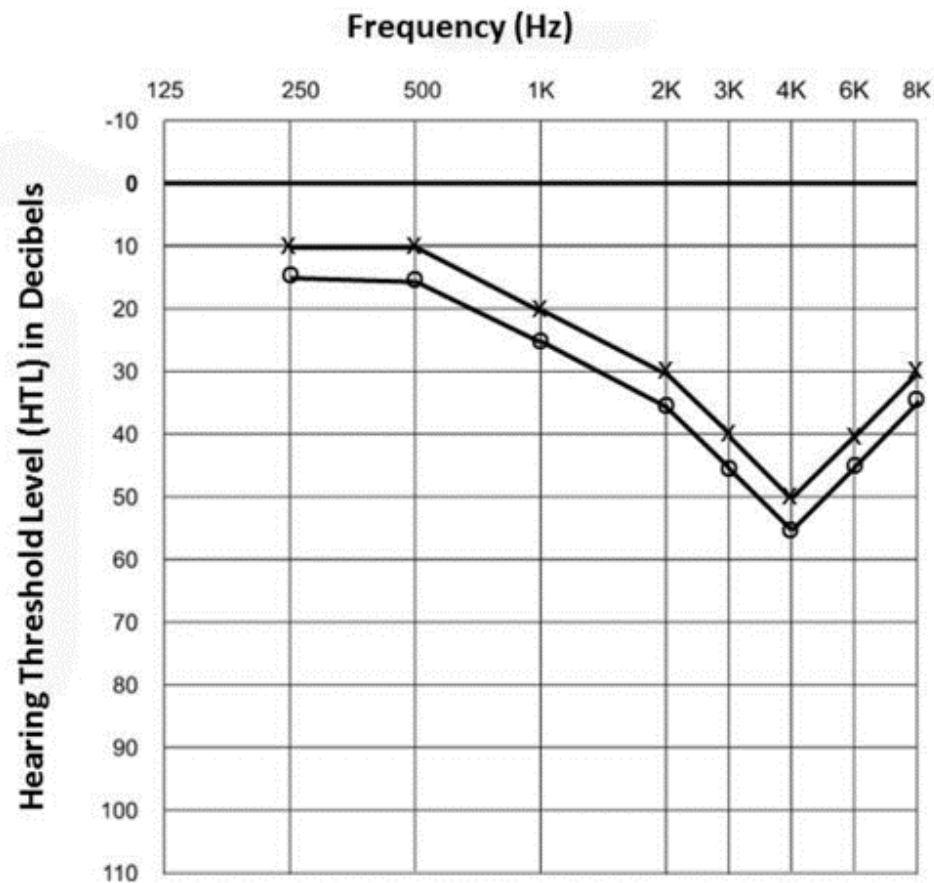
- Objectives
  - obtain a worker's baseline upon work entry
  - provide a working record of a worker's hearing acuity
  - determine effectiveness of noise control measures
  - compliance with regulations

# Industrial Audiometry (cont.)

- Threshold Audiometry
  - measures the threshold levels at which a sound level can be heard at various frequencies
  - compared to a standard reference
  - NIHL most pronounced at 4000 Hz denoted by audiogram

# Industrial Audiometry (cont.)

Sensorineural Hearing Loss Audiogram





# Noise Exposure Regulations

- OSHA Noise Permissible Exposure Level (PEL)
  - 90 dBA for 8 hours
  - hearing conservation program for workers exposed at 85 dBA and above (8-hour TWA)
    - monitoring
    - audiometric testing
    - hearing protection
    - employee training
    - record keeping

# Noise Exposure Regulations (cont.)

- Standard Threshold Shift (STS) as Defined by OSHA
  - an average shift of 10 dB or more in at least one ear
  - average shift at 2000 Hz, 3000Hz, and 4000 Hz
  - current audiogram is compared with baseline
  - allows for age correction factors